



Parameters of Microring modulators.



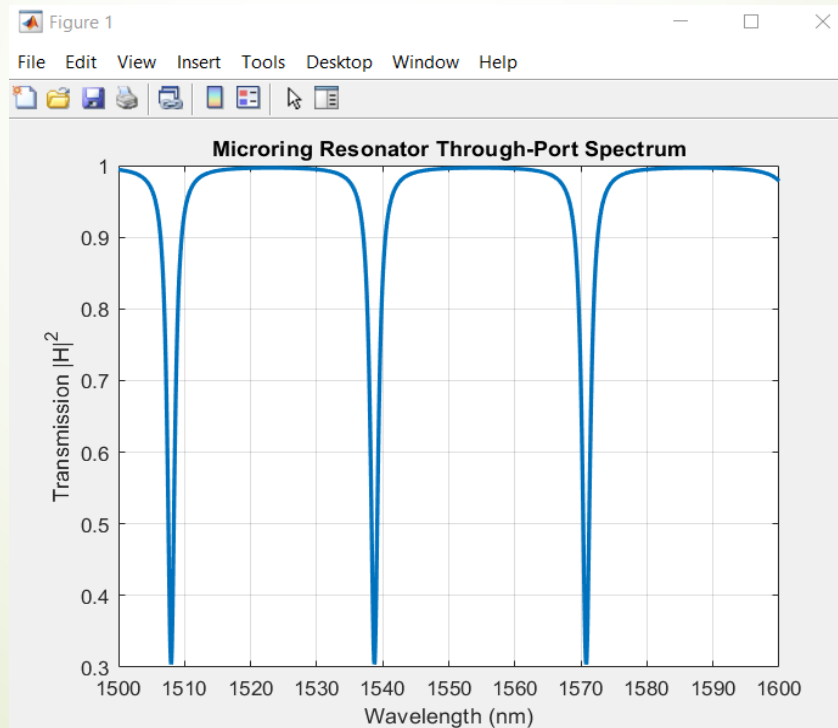
contents

- FSR
- FWHM
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Microring modulators

Free Spectral Range (FSR)

- The separation of successive resonances is termed the FSR.



Microring modulators

Free Spectral Range (FSR)

- At resonance, $\phi = \frac{2\pi n g L}{\lambda} = 2\pi m$ $\lambda = \frac{n g L}{m}$

where L is the round-trip length and m is an integer. Two successive resonances, ω_1 and ω_2 , are then related as

$$\begin{aligned}\text{FSR}_{\text{frequency}} &= \omega_2 - \omega_1 = \frac{2\pi}{T_R} = \frac{2\pi\nu}{L} \\ &= \frac{2\pi c}{n g L} = 2\pi(df)\end{aligned}$$

Translating to wavelength,

$$f = \frac{c}{\lambda} \qquad \frac{df}{d\lambda} = \frac{c}{\lambda^2} \qquad \text{FSR}_{\text{wavelength}} = \frac{\lambda^2}{n g L}$$

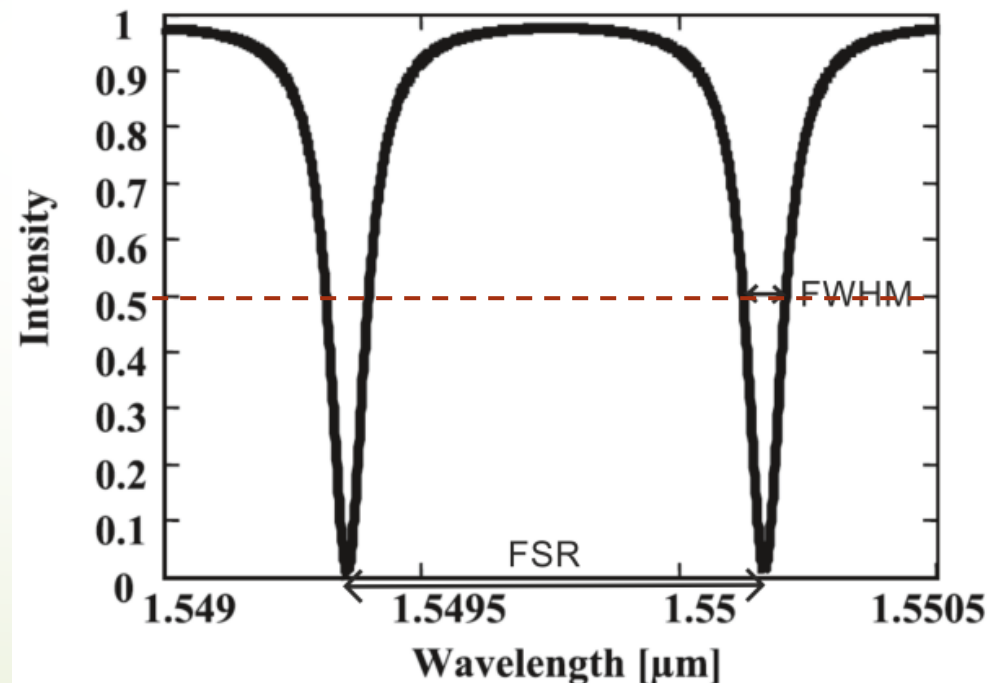
Larger FSR avoids overlap between adjacent channels. Useful in dense wavelength division multiplexing (DWDM), Fewer modes, easier channel isolation.

Smaller FSR gives more resonances in a given spectral window, increasing sensitivity and resolution, but Larger footprint as it depends on the optical path length.

Microring modulators

full width half maximum (**FWHM**)

- the **resonance width** which is analogous to **3dB bandwidth**** of the spectrum in decibel scale.



the **frequency or wavelength range over which the signal power drops to **half** its maximum value.

Microring modulators

Q-factor

-It gives information about the resonance with respect to its central frequency and is given by the ratio of the operating wavelength divided by the resonance width.

$$Q = \frac{\lambda_{res}}{FWHM}$$

-It is a measure of the temporal confinement of light inside the ring resonator and how sharp resonance is.

-It measures the wavelength selectivity of the ring resonator.

Microring modulators

Q-factor

$$Q = \frac{\lambda_{res}}{FWHM}$$

- It is required to obtain narrow FWHM around λ_{res} to achieve high Q-factor.
- This is very useful for wavelength selective filtering characteristics.
- **Lower propagation losses (2.7 dB/cm) allow Q to remain high even at longer lengths, while higher losses (10 dB/cm) cause Q to drop rapidly**

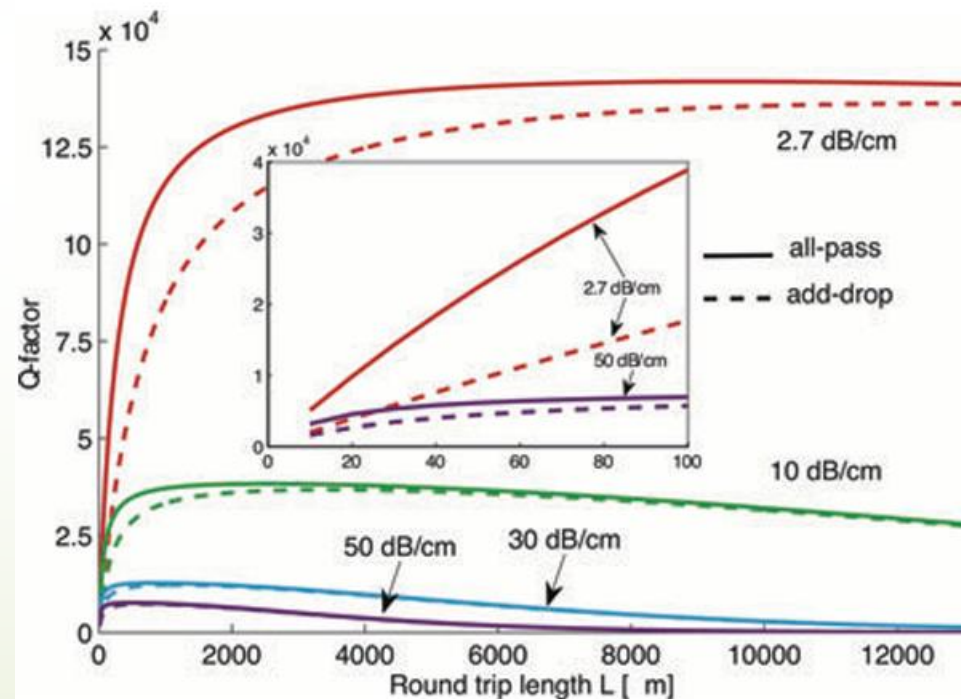


Figure 5 (online color at: www.lpr-journal.org) Quality factor as a function of cavity length for four propagation loss figures.

Microring modulators

Finesse

-The ratio of free spectral range and FWHM or the resonance width of the spectrum, which gives information about the resonance with respect to spacing and is given by:

$$F = \frac{FSR}{FWHM}$$

- Finesse decreases with increasing cavity length, as FSR decreases as cavity length increases.
- These curves show how critical low-loss waveguides are for achieving high-performance ring resonators, especially at longer cavity lengths.

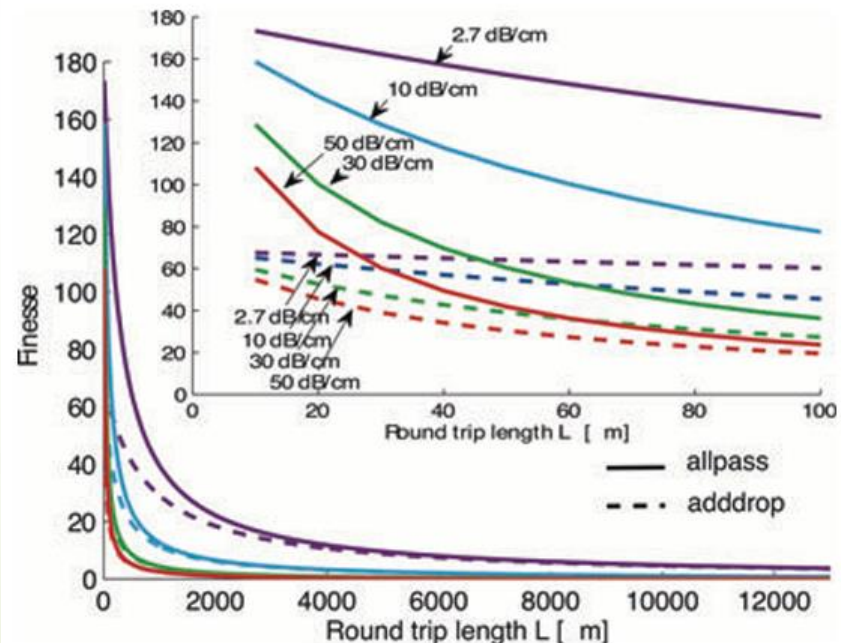


Figure 6 (online color at: www.lpr-journal.org) Finesse as a function of cavity length for four propagation loss figures.

Microring modulators

Group index N_g

-The group index takes into account the dispersion of the wave guide and is defined by

$$n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda}$$

- - The group index, and the corresponding group velocity $v_g = \frac{c}{n_g}$, describes the velocity at which the envelope of a propagating pulse travels and is a characteristic of a dispersive medium (or waveguide).

Microring modulators

Group index Ng

- If the wave is not monochromatic, the wave packet doesn't travel at the velocity of the individual waves but moves with a group velocity

$$v_g = \frac{d\omega}{d\beta} \qquad v_p = \frac{\omega}{\beta}$$

$$\omega = 2\pi f = 2\pi \frac{c}{\lambda}$$

$$\frac{\partial \omega}{\partial \lambda} = -2\pi c \frac{1}{\lambda^2} = -\frac{1}{\lambda} \frac{2\pi c}{\lambda} = -\frac{\omega}{\lambda}$$

Microring modulators

Group index N_g

$$\begin{aligned}\beta &= n_1 \frac{2\pi}{\lambda} = n_1 \frac{\omega}{c} \quad \text{and } v_p = \frac{c}{n_1} \\ v_g &= \frac{\partial \omega}{\partial \beta} = \frac{\delta \lambda}{\delta \beta} \frac{\partial \omega}{\partial \lambda} = \left[\frac{\partial}{\partial \lambda} \left(n_1 \frac{\pi}{\lambda} \right)^{-1} \right] \left(-\frac{\omega}{\lambda} \right) \\ &= \frac{-\omega}{2\pi\lambda} = \left[\frac{1}{\lambda} \frac{dn_1}{d\lambda} - \frac{n_1}{\lambda^2} \right] \\ &= \frac{c}{n_1 - \lambda \frac{dn_1}{d\lambda}} = \frac{c}{N_g}\end{aligned}$$

where $N_g = n_1 - \lambda \frac{dn_1}{d\lambda}$ = group index of the guide

References

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